

pattern that lets them make sense of this treasure trove of data. The challenge here is to understand how to federate it, that is, how integration takes place.

Machine-to-machine device management service providers look at figuring out which among these enormous data are useful and how to segregate it into consumable chunks that can be sold to different organisations or enterprises. The approach they take is to develop an adaptation for each of these machines that communicate in different languages, and hence shield that chaos from the consumer, using an intelligent gateway that presents a unified, sensible view.

There are many institutes, companies and ventures that have been set-up to crack this. University of Nottingham is one such example that recently received a £ 359,000 funding from the UK's innovation agency for its value enhancement for data from assets & transactions (VEDAT) project. The objective of this project is to identify value from high volumes of complex data generated in real-time, through telematics technology—value that can be unlocked by other organisations for various fields like transportation, automation and medicine, to name a few.

Let us take a look at the different sectors that the IoT and Big Data have had an effect on.

... Enabling intelligent manufacturing

Manufacturers using the IoT is nothing new. Semiconductor firm Fujitsu grows their Kirei Yasai lettuce in a facility previously used to fabricate microchips. These lettuce do not just enjoy growing-up in a super-clean location, these are farmed by data-driven agricultural technology, powered by sensor arrays working in tandem with cloud-computing solutions to crunch data and ensure that the environment remains at optimal levels throughout the growth process. This includes monitoring

Fundamentals: Big Data and the IoT

There are roughly three distinct stages for the Internet of Things (IoT). First, data is collected using sensors. At the next step, this data is analysed with the help of complex algorithms that were embedded into the IoT device or cloud based data processing. This is followed by decision-making (using analytical engines) and transmission of data to the decision-making server. If the information collected is quite large and complex that it becomes difficult to analyse using traditional data-processing techniques, we call it Big Data. Results made from this analysis are then transmitted to the actuator system, where the decision is implemented.

temperature, humidity and fertiliser composition as they try to figure out the best growing conditions and ways to control micro-organisms.

Projects like these use solutions like Akisai, a food and agricultural cloud solution that leverages information and communications technology (ICT) to dramatically improve the efficiency of agricultural operations. Akisai will be used to continually analyse agricultural data for highly-productive cultivation and facilitate the entire management process, including management, production and sales, for a more efficient agricultural operation.

Another example of this, as mentioned in the Lopez Research report on building smarter manufacturing with the IoT, is Harley-Davidson's use of the IoT in its York Pennsylvania motorcycle plant. Software in the plant keeps record of how different equipment perform, such as the speed of fans in the painting booth. The software then automatically adjusts the machinery when it detects that the fan-speed, temperature or humidity has deviated from acceptable ranges.

Bosch, GE and Johnson Controls are some companies who are firm believers of Industry 4.0, where robots and other connected machine elements in a factory talk to each other to figure out the most optimal way to do something. This is no easy task.

Solving challenges: Data-collection in a manufacturing environment. Connecting elements in a business together and into cyberspace is no simple thing, and that is

where firms like Rockwell Automation, along with Big Data analytics firms like Computer Systems Corp, come into play. When an application involves a large number of sensor networks that are sending data on continuous basis, it becomes important to capture that information effectively and keep as much of it for further analysis. Such analyses are important to understand trends, forecast events and other functions. In these cases, data size would grow non-linearly in a rather short period of time. In T. Anand's opinion, such applications incorporating Big Data could be useful for manufacturers having deployed the IoT in a full-fledged manner.

To ensure delivery of data collected from factory-wide IoT implementation, manufacturers need networks that can cope with RF challenges in the plant, harsh environmental conditions and reliability for transmission of alarms and real-time data-stream processing. Vendors like Cisco or Emerson Network Power might provide this service. For example, the Lopez research report mentions that General Motors implemented a standard based network architecture, called plant floor controls network (PFCN), to standardise the design of each plant network and establish a single engineering team that monitors and troubleshoots network operations globally.

An alternative to transmitting huge amounts of unprocessed data over factory floors is to have the processing done in the device itself.